أكاديمية طويق
Tuwaiq Academy

Amazon Database System

Tuwaiq Academy Bootcamp: Web development using java & AI integration

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1. Project Brief

This database is designed to simulate a simplified e-commerce system inspired by Amazon. It consists of five tables that manage the core operations of an online store.

- The **customers** table stores user information such as name, age, email, phone, balance, and gender, ensuring each customer has a unique identity.

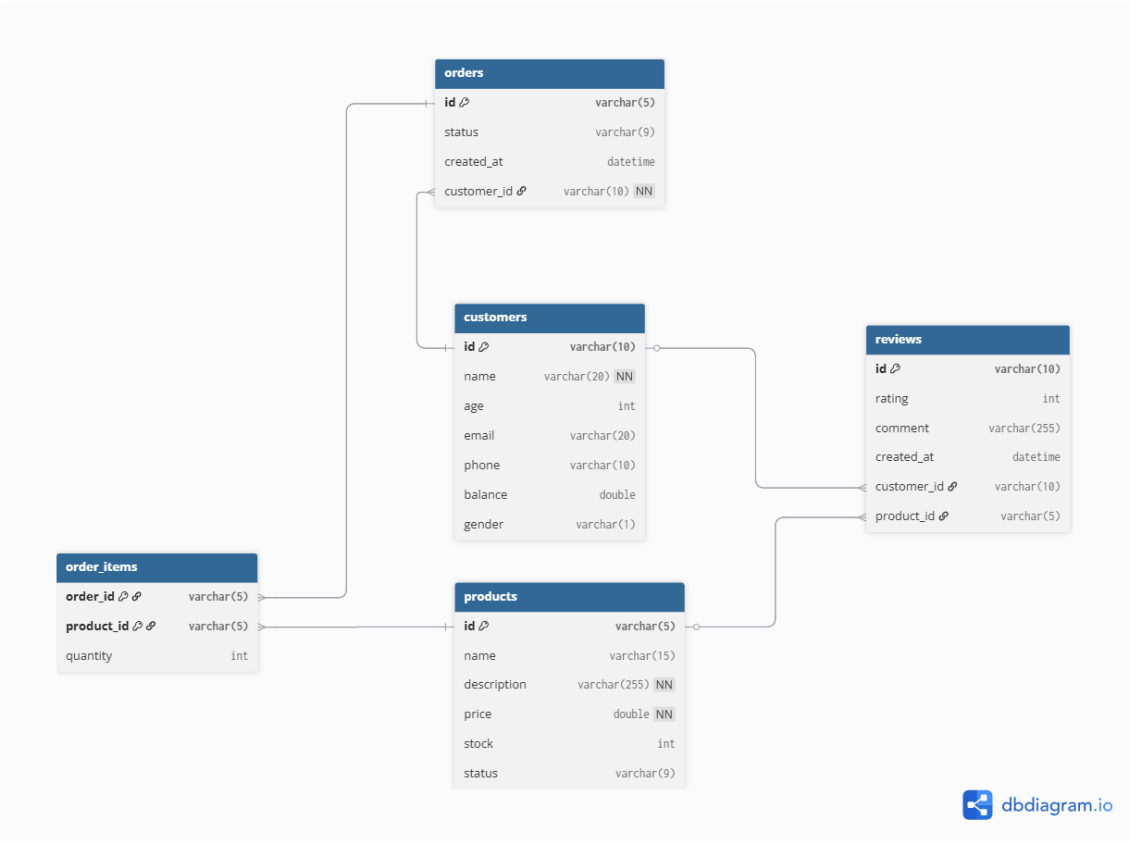
- The **products** table holds all available items in the store, including their description, price, stock quantity, and availability status, which can be either available or sold out.

- The **orders** table tracks each purchase made by a customer, recording the order status whether it is pending or delivered, along with the timestamp of when it was created.

- The **order_items** table stores the details of each order, including which products were included and the quantity of each item purchased.

- The **reviews** table allow customers to leave ratings between 1 and 5 and written comments on products, helping other users make informed purchasing decisions.

2. Database Schema



```sql
#Create database
create database amazon;

#Crate tables
create table customers (
   id varchar(10) primary key ,
   name varchar(20) not null ,
   age int check ( age > 18 ),
   email varchar(20) unique ,
   phone varchar(10) unique ,
   balance double default (0),
   gender varchar(1) check ( gender='M' or gender='F')
);

create table products (
   id varchar(5) primary key ,
   name varchar(15) unique ,
   description varchar(255) not null ,
   price double not null ,
   stock int default (0),
   status varchar(9) check ( status='AVAILABLE' or status='SOLDOUT')
);

create table orders (
   id varchar(5) primary key ,
   status varchar(9) check ( status='PENDING' or status='DELIVERED'),
   created_at datetime default current_timestamp,
   customer_id varchar(10),
   foreign key (customer_id) references customers(id) on delete cascade
);

create table order_items (
   order_id varchar(5),
   product_id varchar(5),
   quantity int default 1,
   primary key (order_id, product_id),
   foreign key (order_id) references orders(id) on delete cascade ,
   foreign key (product_id) references products(id)
);

create table reviews (
  id varchar(10) primary key ,
  rating int check ( rating between 1 and 5),
  comment varchar(255),
  created_at datetime default current_timestamp,
  customer_id varchar(10),
  product_id varchar(5),
  foreign key (customer_id) references customers(id) on delete cascade ,
  foreign key (product_id) references products(id) on delete cascade
);
```

#POST

```
insert into customers values ('C001', 'Ahmed Ali', 25, 'ahmed@mail.com', '0501234567', 5000, 'M');
insert into customers values ('C002', 'Sara Hassan', 30, 'sara@mail.com', '0502345678', 3000, 'F');
insert into customers values ('C003', 'Khalid Omar', 22, 'khalid@mail.com', '0503456789', 1500, 'M');
insert into customers values ('C004', 'Maha Nasser', 28, 'maha@mail.com', '0504567890', 7000, 'F');
insert into customers values ('C005', 'Faisal Saad', 35, 'faisal@mail.com', '0505678901', 2000, 'M');

insert into products values ('P001', 'Laptop', 'High performance laptop', 3000, 50, 'AVAILABLE');
insert into products values ('P002', 'Phone', 'Smartphone', 1500, 100, 'AVAILABLE');
insert into products values ('P003', 'Tablet', 'Tablet device', 900, 30, 'AVAILABLE');
insert into products values ('P004', 'Keyboard', 'New Keyboard', 200, 0, 'SOLDOUT');
insert into products values ('P005', 'Headphones', 'Noise cancelling', 400, 75, 'AVAILABLE');

insert into orders (id, status, customer_id) values ('O001', 'PENDING', 'C001');
insert into orders (id, status, customer_id) values ('O002', 'DELIVERED', 'C002');
insert into orders (id, status, customer_id) values ('O003', 'PENDING', 'C003');
insert into orders (id, status, customer_id) values ('O004', 'DELIVERED', 'C004');
insert into orders (id, status, customer_id) values ('O005', 'PENDING', 'C005');

insert into order_items values ('O001', 'P001', 1);
insert into order_items values ('O002', 'P002', 2);
insert into order_items values ('O003', 'P003', 1);
insert into order_items values ('O004', 'P005', 3);
insert into order_items values ('O005', 'P003', 2);

insert into reviews values ('R001', 5, 'Great laptop', now(), 'C001', 'P001');
insert into reviews values ('R002', 4, 'Good phone', now(), 'C002', 'P002');
insert into reviews values ('R003', 3, 'Cool', now(), 'C003', 'P003');
insert into reviews values ('R004', 2, 'Not worth it', now(), 'C004', 'P005');
insert into reviews values ('R005', 5, 'Like it', now(), 'C005', 'P003');
```

#GET

```
select * from customers;
select * from products;
select * from orders;
select * from order_items;
select * from reviews;

select id, name, email from customers;
select id, name, price, status from products;
select id, status, created_at from orders;

select * from customers where gender = 'M';
select * from customers where balance > 3000;
select * from customers where age < 30;
select * from products where status = 'SOLDOUT';
select * from products where price < 1000;
select * from orders where status = 'DELIVERED';
```

```
select * from reviews where rating < 3;
select * from order_items where quantity > 1;
select * from customers order by balance desc;
select * from products order by price asc;
select * from reviews order by rating desc;
select count(*) as total_customers from customers;
select avg(price) as avg_price from products;
select sum(balance) as total_balance from customers;
select sum(stock) as total_stock from products;
select max(price) as most_expensive from products;


#PUT
update customers set balance = 4500 where id = 'C001';
update customers set email = 'sara.new@mail.com' where id = 'C002';

update products set price = 2800 where id = 'P001';
update products set status = 'SOLDOUT' where id = 'P002';

update orders set status = 'DELIVERED' where id = 'O001';
update orders set status = 'DELIVERED' where id = 'O003';

update order_items set quantity = 3 where order_id = 'O001' and product_id = 'P001';
update order_items set quantity = 1 where order_id = 'O002' and product_id = 'P002';

update reviews set rating = 4 where id = 'R003';
update reviews set comment = 'Amazing product!' where id = 'R005';


#DELETE
delete from reviews where id = 'R004';
delete from order_items where order_id = 'O004' and product_id = 'P005';
delete from orders where id = 'O005';
delete from products where id = 'P004';
delete from customers where id = 'C005';
```